MTH309 PRACTICE PROBLEMS WEEK 13

Question 1. Let X and Y be two real valued, independent random variables defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that $\mathbb{P}(X+Y=a)=1$ some real number a. Show that both X and Y are constant random variables.

Question 2. Let $F, F_1, F_2, \cdots$ be the distribution functions corresponding to the probability measures $\mu, \mu_1, \mu_2, \cdots$ on $(\mathbb{R}, \mathbb{B}_{\mathbb{R}})$. Prove the equivalence between the following two statements.

(1) $\lim_n F_n(x) = F(x)$, for all continuity points x of F, where $F_n(y) := \mu_n((-\infty, y]), F(y) := \mu((-\infty, y])$, $\forall y \in \mathbb{R}$ (corresponding distribution functions).
(2) There exists a dense set D in $\mathbb{R}$ such that $\lim_n F_n(x) = F(x), \forall x \in D$.

Question 3. Compute the Characteristic function for a Cauchy random variable and check that it is not differentiable at 0. (Note that the first absolute moment does not exist for a Cauchy distribution).

Question 4. Let $\{X_n\}_n$ be a sequence of i.i.d. Cauchy RVs defined on the same probability space. Does some LLN hold for this sequence?

Question 5. Show that the Characteristic function of an RV is uniformly continuous.

Question 6. Let Φ_X be the Characteristic function of an RV X. Construct a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and RVs Y and Z on this space, such that $\Phi_Y = \overline{\Phi}_X$ (the complex conjugate) and $\Phi_Z = |\Phi_X|^2$.

Question 7. Let X be an n-dimensional random vector, $a \in \mathbb{R}^m$ and $A = (a_{ij})$ be an $m \times n$ real matrix. Recall that the Characteristic function of X is given by

$$\Phi_X(u) = \mathbb{E}\exp(iu^t X), \forall u \in \mathbb{R}^n.$$

Show that the Characteristic function of $Y = AX + a$ is given by

$$\Phi_Y(u) = \exp(iu^t a)\Phi_X(A^t u), \forall u \in \mathbb{R}^n.$$